Day 04

Packet Capture

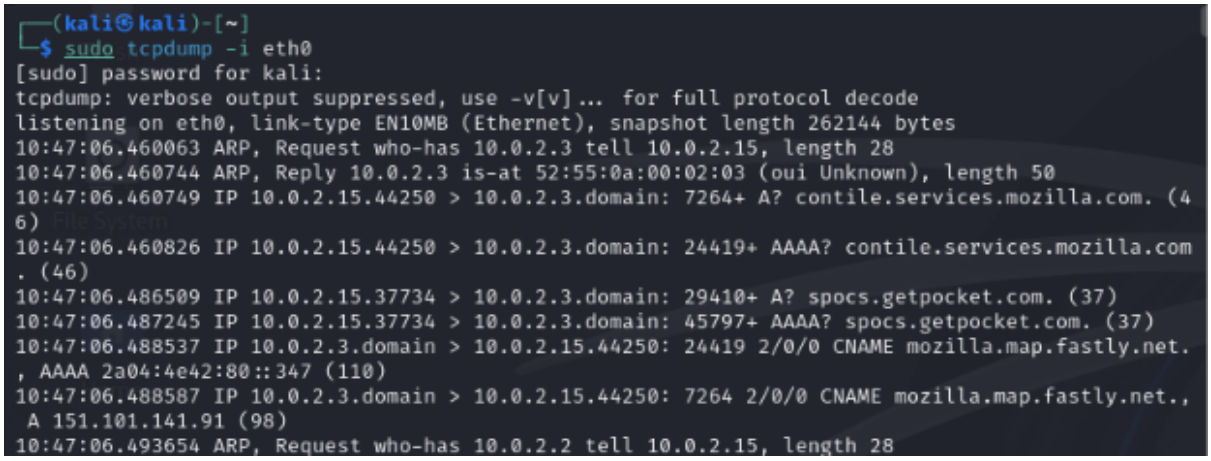


Figure 1: sudo tcpdump -i eth0

Captures and displays the network packets travelling through the eth0 interface.

Wireshark

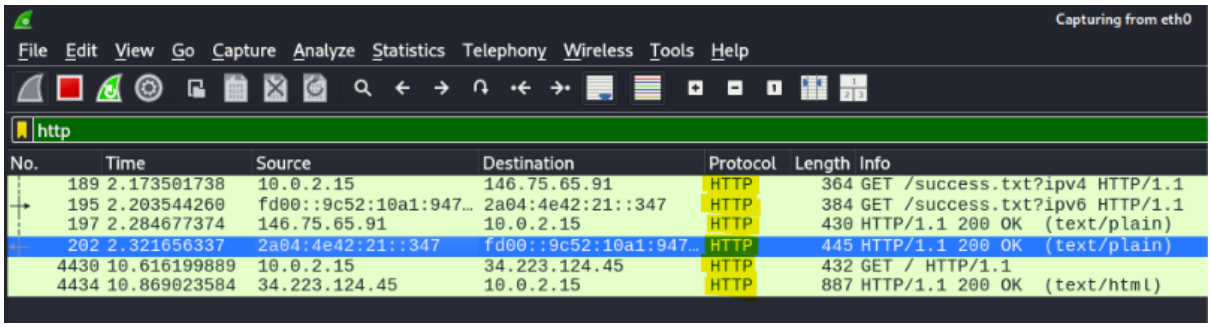


Figure 2a: Captured HTTP

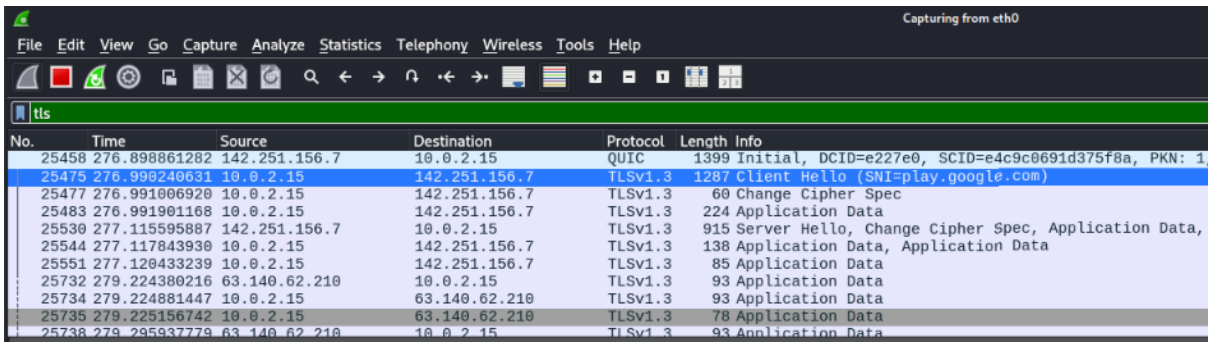


Figure 2b: Captured HTTPS

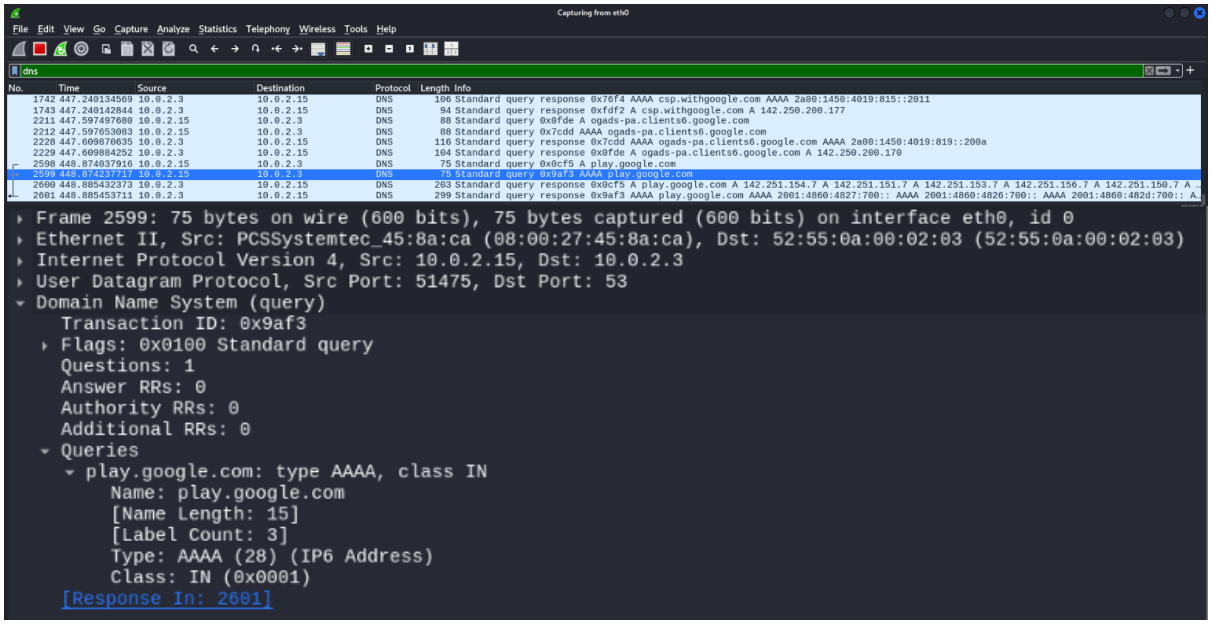


Figure 3: Observed DNS queries

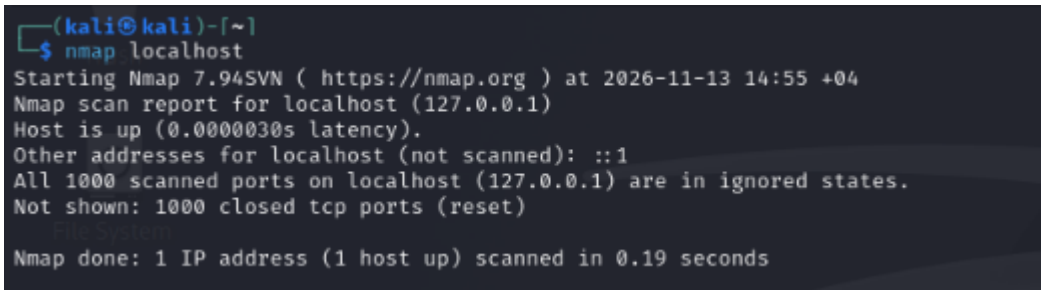


Figure 4: nmap localhost

Scans localhost to identify open ports and discover which services are listening on them.

Hands on Task

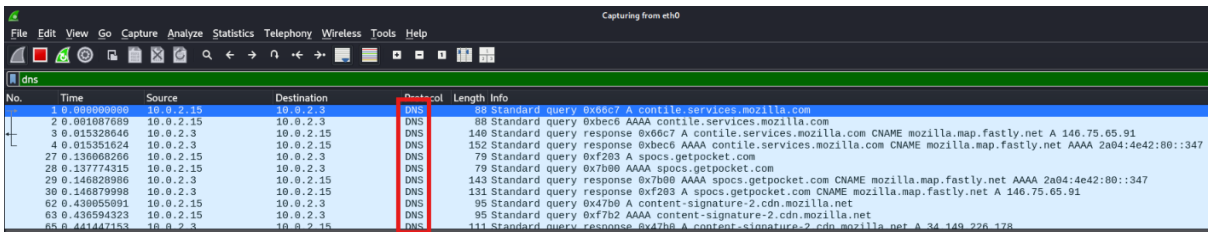


Figure 5: Identified DNS requests

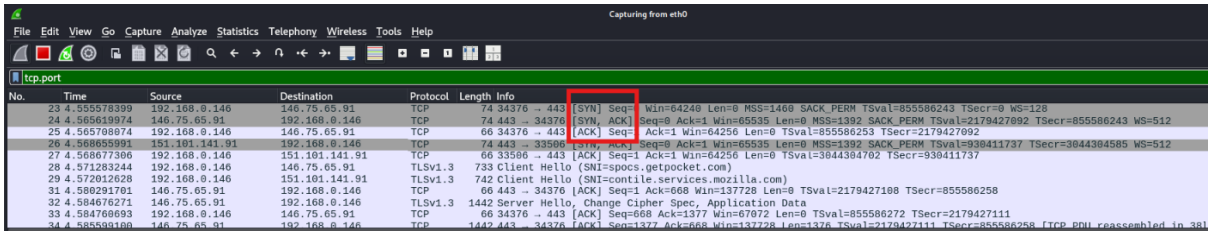


Figure 6: Identified TCP handshake.

HTTPS is secure because it uses TLS (transport layer security) encryption to protect the data exchanged between a browser and a web server. It encrypts information so that attackers who intercept the network traffic can't easily read or understand its contents. TLS also provides authentication by verifying the servers digital certificate, helping to make sure that users are communicating with the intended website. It protects the data integrity, helping prevent information from being modified during transmission.